Md Nafiu Rahman

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About Me

I am a Lecturer in Computer Science and Engineering at BRAC University and a software engineering researcher. My work applies large language models to software security, such as detecting leaked secrets in source code and issue reports, and studies whether tests actually check the behavior they cover. My long-term goal is to build intelligent tools that improve developer productivity and software reliability. I am seeking PhD positions starting in Fall 2027.

Education

Bangladesh University of Engineering & Technology (BUET), Dhaka *Feb 2020 – Mar 2025*
B.Sc. in Computer Science & Engineering CGPA: 3.81 / 4.00 (Final year: 3.94 / 4.00)

- Relevant coursework: Data Structures & Algorithms, Object Oriented Programming, Software Engineering, Database Systems, Numerical Methods, Machine Learning.

Notre Dame College, Dhaka *Jan 2017 – Jan 2019*
Higher Secondary Certificate GPA: 5.00 / 5.00

Bir Shrestha Noor Mohammad Public College, Dhaka *Jan 2009 – Dec 2016*
Secondary School Certificate GPA: 5.00 / 5.00

Work Experience

Summer Research Intern, University of Illinois Urbana-Champaign (UIUC), Urbana-Champaign, IL
May 2026 – July 2026
UIUC++ Summer Research in Software Engineering (SRSE) 2026
Supervisor: Dr. Shin Hwei Tan

Lecturer, Department of Computer Science and Engineering, BRAC University, Dhaka July 2025 – Present

- Teaching undergraduate courses including *Numerical Methods (CSE330)* and *Software Engineering (CSE470)*, delivering theory and laboratory sessions.
- Designing course materials, programming assignments, laboratory exercises, examinations, and coordinate lab activities while mentoring students in software engineering best practices.

Publications

* *Equal contribution.*

Conference Papers

[C1] An Empirical Study of Security Calibration in Large Language Models for Code.
Authors: Mohammed Latif Siddiq, **Md Nafiu Rahman**, Joanna C. S. Santos.
(ICSME 2026 Research Track)
Accepted at ICSME 2026 Research Track

[C2] Secret Breach Detection in Source Code with Large Language Models.
Authors: **Md Nafiu Rahman***, Sadif Ahmed*, Zahin Wahab*, S. M. Sohan, Rifat Shahriyar.
(ESEM 2025 Technical Track)
Accepted at ESEM 2025 Technical Track

[C3] Secret Leak Detection in Software Issue Reports using LLMs: A Comprehensive Evaluation.
Authors: Sadif Ahmed*, **Md Nafiu Rahman***, Zahin Wahab*, Gias Uddin, Rifat Shahriyar.
(MSR 2026 Technical Track)
Accepted at MSR 2026 Technical Track

[C4] IssueGuard: Real-Time Secret Leak Prevention Tool for GitHub Issue Reports.
Authors: **Md Nafiu Rahman***, Sadif Ahmed*, Zahin Wahab, Gias Uddin, Rifat Shahriyar.
(FSE 2026 Tool Demonstration Track)
Accepted at FSE 2026 Tool Demonstration Track

Workshop Papers

[W1] BanglaForge: LLM Collaboration with Self-Refinement for Bangla Code Generation.
Authors: Mahir Labib Dihan, Sadif Ahmed, **Md Nafiu Rahman**.
(BLP Workshop at AACL-IJCNLP 2025)
Accepted at BLP Workshop, AACL-IJCNLP 2025

Journal Papers

[J1] A Survey on Agentic Security: Applications, Threats and Defenses.
Authors: Asif Shahriar*, **Md Nafiu Rahman***, Sadif Ahmed*, Farig Sadeque, Md Rizwan Parvez.
(Transactions on Machine Learning Research, 2026)
Accepted at Transactions on Machine Learning Research (TMLR)

Manuscripts Under Review

[P1] Covered Is Not Checked: Which Branch Outcomes Would Your Tests Notice?
Authors: **Md Nafiu Rahman***, Cemil Canbaz*, Soneya Binta Hossain.
(arXiv)

[P2] AgentTell: Behavioural Side-Channel Leakage in Browser-Use Agents.
Authors: Asif Shahriar*, **Md Nafiu Rahman***, Sadif Ahmed*, Farig Sadeque, Md Rizwan Parvez.
(arXiv)

[P3] From Codebases to LLMs: Non-Inclusive Naming in Linux Foundation Repositories .
Authors: Honghao Tan, **Md Nafiu Rahman**, Shin Hwei Tan.
(arXiv)

[P4] Explainable Transformer-CNN Hybrid for Modeling Brain Aging from MRI Images.
Authors: Wasif Jalal, **Md Nafiu Rahman**, Md Sohel Rahman.
(arXiv)

Preprints

[P5] EVCC: Enhanced Vision Transformer-ConvNeXt-CoAtNet Fusion with Adaptive Routing for Classification.
Authors: Kazi Reyazul Hasan, **Md Nafiu Rahman**, Sadif Ahmed, Wasif Jalal, Shahriar Raj, Mubasshira Musarrat, Muhammad Abdullah Adnan.
(arXiv)

Services

Reviewer

- Reviewer, ACL Rolling Review - May 2026

Organizer

- BUET CSE FEST, 2024

Talks

- *Secret Leak Detection in Software Issue Reports using LLMs: A Comprehensive Evaluation*, MSR 2026 – Video
- *IssueGuard: Real-Time Secret Leak Prevention Tool for GitHub Issue Reports*, FSE 2026 Tool Demonstration Track – Video
- *Secret Breach Detection in Source Code with Large Language Models*, ESEM 2025 – Video

Technical Skills

Data Science & ML: Python, NumPy, Pandas, Scikit-learn, PyTorch, TensorFlow, Torchvision.
Databases: PostgreSQL, PL/pgSQL, MongoDB, Firebase Firestore.
Full-Stack: Node.js (backend), HTML/CSS, React, Svelte, Flutter.
Languages: C/C++, Python, Java, JavaScript / TypeScript, PHP, Bash, Dart.

Language Proficiency

Bengali Native
English IELTS Overall 8.0 – Listening: 9, Reading: 9, Writing: 7, Speaking: 6.5

Achievements

- Top 20 finalists — Robi Datathon 2024 (national deep learning competition).
- Dean's list award and university merit scholarship at BUET.
- 5th at BLP 2025 Code Generation Challenge

Academic Projects

- **Machine Learning Algorithms and Neural Network from Scratch** – github.com/nafiurahman00/CSE-472 – Implementations of core ML algorithms (logistic regression with ensembles, PCA/SVD for reconstruction, EM clustering) and a feed-forward neural network with Adam optimizer built from numpy.
- **Cryptography and Security Attacks** – github.com/nafiurahman00/CSE-406 – Implemented AES encryption and Diffie-Hellman key exchange, socket communication demos, and reviewed a mobile pentesting framework as part of course project.
- **Network Simulation** – github.com/nafiurahman00/CSE-322 – Implemented Congestion Control Algorithm, threaded server-client sockets, error correction algorithms and simulated wired/wireless mobility scenarios.
- **Operating System Internals with xv6** – github.com/nafiurahman00/CSE-314 – Implemented threading and synchronization primitives, system calls, and explored scheduler internals in xv6.
- **BusBuddy (Android)** – github.com/nafiurahman00/BusBuddy-Client-End – Flutter app with Node.js backend, PostgreSQL and Firebase integration. Provided ticketing, schedules, tracking and real-time updates for university bus users.
- **Nishorgo (E-commerce)** – Term-Project-2-2-Nishorgo – Full stack e-commerce site for plant sales with filtering, cart, admin analytics and order management.
- **Compiler (subset of C)** – github.com/nafiurahman00/Compiler – Subset-of-C compiler using Lex/Yacc and 8086-style assembly generation: lexer, parser, and intermediate code generation.
- **Catch the Egg (Game)** – github.com/nafiurahman00/Catch-The-Egg – OpenGL / Igraphics game for catching falling eggs; implemented game mechanics, scoring and difficulty scaling.

References

- **Dr. Rifat Shahriyar**, Professor, Department of CSE, Bangladesh University of Engineering and Technology (BUET) – rifat.shahriyar@gmail.com – rifat@cse.buet.ac.bd
- **S. M. Sohan**, Senior Engineer, Google – sohan39@gmail.com
- **Dr. Md Rizwan Parvez**, Scientist, Qatar Computing Research Institute (QCRI) – rizwan@ucla.edu – rizwan.incipient@gmail.com
- **Dr. Farig Sadeque**, Associate Professor, Department of CSE, BRAC University – farig.sadeque@bracu.ac.bd